MICHIGAN MEDICINE
UNIVERSITY OF MICHIGAN

PEDIATRIC PROCALCITONIN USAGE GUIDELINES

Recommendations:

- *Procalcitonin does not meaningfully inform the use of antibiotics in most scenarios in which an appropriate diagnostic assessment is taken, and guideline-concordant antibiotic selection and duration of therapy is utilized*.
- Procalcitonin testing IS indicated in the assessment of febrile infants aged 3-60 days
    - Febrile Infant Pathway: *https://michmed-clinical.policystat.com/policy/12261468/latest/*
    - Results should be utilized according to the febrile infant pathway
- Procalcitonin testing may be *considered* in:
    - Patients with possible bacterial pneumonia based on clinical and radiographic assessment AND
    - With diagnostic uncertainty (i.e., alternative explanation for respiratory symptoms, questionable infiltrates, possible viral etiology) **AND**
    - Without characteristics that may result in false-positive results (see below)
- Procalcitonin is NOT recommended in all other patients, including those with
    - High suspicion for bacterial pneumonia
    - Severe sepsis/septic shock
    - Non-respiratory infectious concerns

In candidate patients as above, the following algorithm can be used to make decisions about the diagnosis and treatment of **pneumonia** with antibiotics:

Pediatric Procalcitonin Level (ng/mL)	Recommendation
≤ 0.5	Antibiotics discouraged
> 0.5	Antibiotics can be considered

Comments:

- Procalcitonin should be evaluated in context with all findings and the total clinical status; clinical judgment always necessary. Procalcitonin should not be used in isolation to decide whether to initiate antibiotics in patients with suspected bacterial pneumonia.
- PCT may not be elevated in pneumonia due to atypical bacterial pathogens such as *Mycoplasma pneumoniae* and *Chlamydia pneumoniae*.
- PCT should NOT be used to *extend* treatment duration, and continuation of antibiotics beyond standard durations, in the setting of clinical stability, is NOT recommended, regardless of PCT level.

Background and Rationale

Procalcitonin (PCT) is a 116 amino acid precursor of calcitonin that has both hormonal and cytokine-like activities and is produced in the thyroid gland by C-cells. Serum concentrations of PCT are usually undetectable; it is more likely to be produced in response to bacterial than viral infection. As a result, it has been studied as a biomarker for the diagnosis and monitoring of acute bacterial infections. **However, there are numerous limitations to the use of PCT which hinder its reliability and utility in routine use.**

False positives are numerous (see below). Of special note, PCT levels are increased in CKD and AKI, which are common in hospitalized patients. False negatives (also below) can also occur. PCT has not been extensively studied in pregnant women and in significantly immunocompromised patients.

When may FALSE POSITIVES occur?
- Postpartum women
- Severe trauma or burns
- Major surgery
- Therapeutic cooling after cardiac arrest
- Treatment with agents which stimulate cytokines
- Acute graft vs. host disease
- Non-septic shock causing decreased organ perfusion and/or infarction
- Kidney disease: Patients with CKD and AKI have falsely elevated PCT levels, with increasing disease associated with progressively increased levels.

When may FALSE NEGATIVES occur?
- Early in infection
- Chronic infections (endocarditis, osteomyelitis, prosthetic/graft infections
- Some localized infections (cellulitis, intra-abdominal abscess)
- Infection due to *Chlamydia pneumoniae* and *Mycoplasma pneumonia*

- **Sepsis/Septic shock**: The use of PCT in suspected sepsis/septic shock is limited by the lack of data in significant patient populations (as above), numerous false positives, and that PCT often takes 6-12 hours to rise after an acute bacterial infection and can be falsely low at the beginning. As a result, 2021 Surviving Sepsis guidelines for adults suggest against incorporating PCT into decisions of whether to start antimicrobials, and the 2020 Surviving Sepsis guidelines for children do not discuss the use of PCT in starting antibiotics. The pediatric guidelines discuss but do not provide a recommendation for or against PCT use in antibiotic de-escalation, while the adult guidelines provide a weak recommendation to use PCT AND clinical evaluation to decide when to discontinue antimicrobials in patients with adequate source control where optimal duration is unclear. However, in clinical trials, the average duration of antimicrobial use in the PCT arm was 5 days, which is a reasonable duration for sepsis without an evident source in patients who are responding clinically. Additionally, PCT did not reduce antibiotic duration in pediatric intensive care unit patients (Katz PIDJ 2022). As a result, it is not evident that PCT provides additional benefit to modern practice.
- **Community-acquired pneumonia**: Randomized controlled trials (RCTs) in patients with mostly community-acquired lower respiratory tract infection (LRTI) have demonstrated that strict adherence to PCT algorithms is associated with shorter durations of antibiotic therapy and may result in decreased antibiotic prescribing rate. However, the duration of therapy in the PCT arms were consistent with currently recommended durations of therapy for community-acquired pneumonia (CAP), i.e., 5 days, while control arm durations exceeded such recommended durations. As such, it is not clear that PCT has a role in CAP when treating for guideline-appropriate durations. Indeed, in a randomized trial in the United States of 1,656 adult patients with LRTI, procalcitonin guidance was **not** associated with a reduction in antibiotic-days (4.2 days in procalcitonin group vs. 4.3 days in control group) (Huang NEJM 2018). PCT had a modest impact on reducing initiation of antibiotics in hospitalized children with LRTI (85% vs 100%, Esposito Respiratory Medicine 2011).
- **Protocolized Trials vs. 'Real Life' Use**: The use of PCT has primarily been found to be beneficial (i.e., decreased antibiotic durations with similar clinical outcomes) in RCTs with strict inclusion/exclusion criteria and high adherence to algorithms. Some 'real life' studies have shown availability of PCT to be associated with *increased* antibiotic consumption.
- **Febrile young infants:** PCT has emerged as the most accurate inflammatory marker for risk stratification in febrile young infants at risk of serious bacterial infections, as described in the 2021 AAP CPG for Evaluation and Management of Well-Appearing Febrile Infants.

Last revised:	**Antimicrobial Stewardship Subcommittee:**	**P&T:**

Revision History	
Date	**Revision**
12/2024	Document Creation
01/2025	P&T Approval

The recommendations in this guide are meant to serve as treatment guidelines for use at Michigan Medicine facilities. If you are an individual experiencing a medical emergency, call 911 immediately. These guidelines should not replace a provider's professional medical advice based on clinical judgment, or be used in lieu of an Infectious Diseases consultation when necessary. As a result of ongoing research, practice guidelines may from time to time change. The authors of these guidelines have made all attempts to ensure the accuracy based on current information, however, due to ongoing research, users of these guidelines are strongly encouraged to confirm the information contained within them through an independent source